Classical Mechanics

Chapter 6 — Forces, Energy, and Oscillations

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6.1 The Inclined Plane

Consider a block of mass m resting on a frictionless inclined plane that makes an angle theta with the horizontal. The gravitational force mg acts vertically downward and can be decomposed into two orthogonal components: one parallel to the surface, equal to mg sin(theta), and one perpendicular to the surface, equal to mg cos(theta). The perpendicular component is balanced by the normal reaction force N exerted by the plane.

The unbalanced parallel component accelerates the block down the slope according to Newton's second law: a = g sin(theta). When kinetic friction is present, with coefficient mu, the net acceleration becomes a = g (sin(theta) - mu cos(theta)). This expression yields a critical angle theta_c = arctan(mu) below which the block remains stationary, illustrating how the limiting case of static equilibrium emerges from the dynamic equations.

The energy perspective offers an equivalent description. As the block descends a vertical height h, gravitational potential energy mgh is converted into kinetic energy (1/2) m v^2. On a frictionless slope, conservation of mechanical energy gives v = sqrt(2 g h), independent of the angle of inclination. Friction would dissipate part of that energy as heat, reducing the final speed accordingly.

6.2 The Simple Pendulum and Harmonic Motion

A simple pendulum consists of a point mass suspended from a fixed pivot by a massless string of length L. When displaced by a small angular amplitude theta_0, the restoring component of gravity, equal to -mg sin(theta), is approximately linear in the angle, and the bob undergoes simple harmonic motion. The angular frequency of oscillation is omega = sqrt(g/L), giving a period T = 2 pi sqrt(L/g) that depends only on length and local gravity.

The motion can be described by the differential equation d^2 theta / dt^2 + (g/L) sin(theta) = 0. For small angles the equation linearizes to d^2 theta / dt^2 + (g/L) theta = 0, whose solution is theta(t) = theta_0 cos(omega t + phi). Plotting the angular position over time produces a sinusoid; plotting velocity against position produces an elliptical phase-space trajectory.

Damping introduces a velocity-proportional term and the amplitude decays exponentially as exp(-gamma t). Driven by an external periodic force, the system exhibits resonance at omega = omega_0, where the steady-state amplitude grows dramatically. This phenomenon underlies the design of seismometers, the tuning of musical instruments, and the catastrophic collapse of the Tacoma Narrows bridge in 1940.

6.3 Projectile Motion

A projectile launched with initial speed v_0 at an angle alpha above the horizontal, in the absence of air resistance, follows a parabolic trajectory. Decomposing the velocity into horizontal and vertical components yields $v_x = v_0 \cos(\alpha)$, constant in time, and $v_y(t) = v_0 \sin(\alpha) - g t$. Integrating gives $x(t) = v_0 \cos(\alpha) t$ and $y(t) = v_0 \sin(\alpha) t - (1/2) g t^2$.

The range R, time of flight T, and maximum height H follow directly: $R = v_0^2 \sin(2 \alpha) / g$, $T = 2 v_0 \sin(\alpha) / g$, and $H = v_0^2 \sin^2(\alpha) / (2 g)$. The range is maximized at alpha = 45 degrees on level ground; it decreases symmetrically on either side. Plotting R as a function of alpha for a fixed v_0 produces a sinusoidal curve peaking at the optimal angle.

Real projectiles are subject to drag, which depends nonlinearly on speed. At low Reynolds numbers the drag force scales linearly with velocity, while at high speeds it scales as v^2. The resulting trajectories are no longer parabolic — they become asymmetric, with shortened range and a reduced impact angle compared with the ideal case.